

E-Commerce README FILE

An end-to-end e-commerce analytics solution that transforms raw transactional data into a decision-ready, interactive 5-page Power BI dashboard. The workflow covers data cleaning in Pandas, business analysis in SQL, data modeling, DAX measures, and interactive reporting.

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Overview

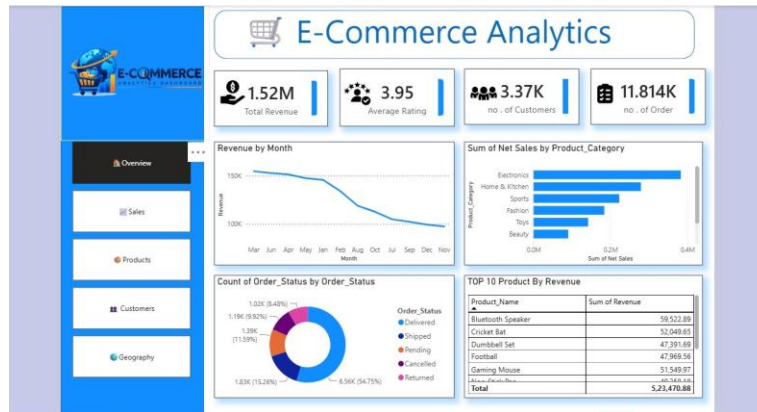
Project	End-to-end E-Commerce Analytics
Tools	Python • Pandas • SQL • Power BI • DAX
Dashboard Pages	Overview • Sales • Products • Customers • Geography
Dataset	12,180 rows • 13 columns

Core Business Questions

- How is revenue changing over time?
 - Which products and categories drive the most value?
 - Who are the highest-value customers?
 - Which payment methods and order statuses matter most?
 - Which countries and cities perform best?
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Dashboard Preview

Overview

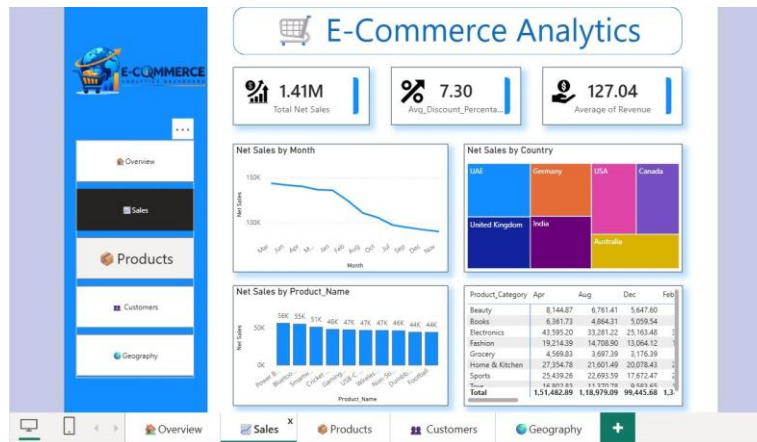


Overview

Key metrics: 1.52M Total Revenue • 3.37K Customers • 11.814K Orders • 3.95 Average Rating

Tracks revenue trend by month, net sales by product category, order status distribution, and the top 10 products by revenue.

Sales

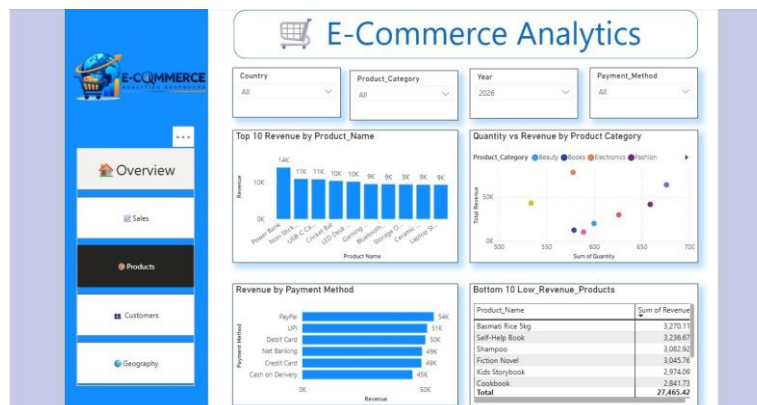


Sales

Key metrics: 1.41M Total Net Sales • 127.04 Average Revenue • 7.30 Avg. Discount %

Breaks down net sales by month, by country, and by product name, plus a monthly product-category matrix.

Products



Products

Interactive filters for Country, Product_Category, Year, and Payment_Method. Shows top 10 revenue products, quantity vs. revenue by category, revenue by payment method, and the bottom 10 low-revenue products.

Customers

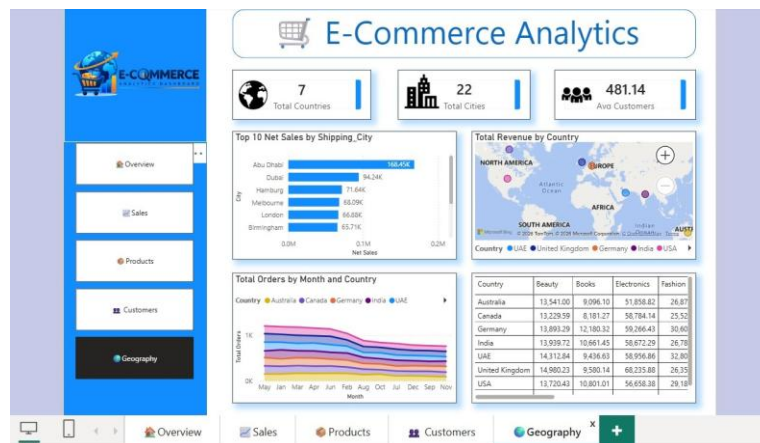


Customers

Key metrics: 3K Total Customers • 3.95 Average Rating • 452.17 Customer Spend

Segments customers (Frequent / High Value / Low Value / Occasional), breaks down orders by status, and surfaces the top 10 customers by revenue.

Geography



Geography

Key metrics: 7 Countries • 22 Cities • 481.14 Avg. Customers

Maps total revenue by country, top 10 net sales by shipping city, and total orders by month and country.

Project Structure

```
E-Commerce-Analytics/  
├── README.md  
├── Dataset/  
│   └── ecommerce_dataset.csv  
├── Python/  
│   └── data_cleaning.ipynb
```

```
├── SQL/
│   └── ecommerce_analysis.sql
├── PowerBI/
│   └── Ecommerce_Analytics.pbix
└── Images/
    ├── overview.png
    ├── sales.png
    ├── products.png
    ├── customers.png
    └── geography.png
```

Tech Stack

- **Python (Pandas)** — data cleaning, standardization, and preprocessing
 - **SQL** — business-question-driven analysis (product, customer, category, payment, geography)
 - **Power BI** — data modeling and interactive 5-page dashboard
 - **DAX** — custom measures for KPIs and calculated metrics
-

Data Cleaning

The raw dataset contained inconsistent date formats, missing values, inconsistent payment/country labels, and invalid negative quantity values that required treatment before analysis.

Missing Values — Before Cleaning

Column	Missing Values
Discount_Percent	735
Shipping_City	502
Customer_Rating	6,150

Cleaning Steps

1. Date conversion & missing-value treatment

```
sales["Order_Date"] = pd.to_datetime(
    sales["Order_Date"], format="mixed"
)

sales["Discount_Percent"].fillna(
    sales["Discount_Percent"].mean(), inplace=True
)

sales["Customer_Rating"].fillna(
    sales["Customer_Rating"].median(), inplace=True
)

sales["Shipping_City"].fillna(
    sales["Shipping_City"].mode()[0], inplace=True
)
```

2. Categorical standardization (Payment Method & Country)

```
sales.loc[:, "Payment_Method"] = sales["Payment_Method"].replace({
    "upi": "UPI", "U.P.I": "UPI",
    "Credit_card": "Credit Card",
    "cod": "Cash on Delivery",
    "NetBanking": "Net Banking",
    "Pay Pal": "PayPal",
    "paypal": "PayPal"
})
```

```

    })

    sales.loc[:, "Country"] = sales["Country"].replace({
        "AU": "Australia", "IN": "India",
        "US": "United States", "USA": "United States",
        "UK": "United Kingdom", "U.K.": "United Kingdom",
        "DE": "Germany", "CA": "Canada",
        "UAE": "United Arab Emirates"
    })

```

3. Invalid data removal Rows with negative `Quantity` values were identified and removed before analysis, since they represent data-entry errors rather than valid returns/refunds.

Missing Values — After Cleaning

Column	Missing Values
Discount_Percent	0
Shipping_City	0
Customer_Rating	0

Note on Customer_Rating: ~50% of rating values were missing, since ratings are typically optional at checkout. Rather than dropping this large a share of rows (which would bias revenue and order-volume metrics), missing ratings were imputed with the median. This preserves the full order history for revenue/sales analysis while flagging rating-based insights as directional rather than exact.

Cleaning Outcome

- `Order_Date` converted to `datetime64[ns]`
- Missing values in targeted fields reduced to zero
- Invalid quantity records removed before analysis
- Payment-method and country labels standardized for consistent Power BI grouping
- Cleaned dataset exported as `cleaned_sales.csv` for downstream analysis

SQL Analysis

SQL analysis supports product, customer, category, payment, order-status, and geographic reporting. The full query set is available in [SQL/ecommerce_analysis.sql](#).

Example — Top 10 Products by Revenue

```

SELECT Product_Name, SUM(Revenue) AS Total_Revenue
FROM Ecommerces
GROUP BY Product_Name
ORDER BY Total_Revenue DESC
LIMIT 10;

```

Core DAX Measures

Measure	DAX
Total Revenue	<code>SUM(Ecommerces[Revenue])</code>
Total Net Sales	<code>SUM(Ecommerces[Net Sales])</code>
Total Orders	<code>COUNT(Ecommerces[Order_ID])</code>
Total Customers	<code>DISTINCTCOUNT(Ecommerces[Customer_ID])</code>

Average Rating	AVERAGE(Ecommerce[Customer_Rating])
Customer Spend	DIVIDE([Total Revenue], [Total Customers], 0)

Business Value

- Identify high-performing products and categories
 - Understand customer value and purchasing behavior
 - Monitor revenue and sales trends over time
 - Compare payment methods and order statuses
 - Identify strong countries and shipping cities
 - Support data-driven commercial decisions
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How to Run This Project

1. Clone the repository

```
git clone https://github.com/<your-username>/E-Commerce-Analytics.git
cd E-Commerce-Analytics
```

2. Data cleaning — open and run Python/data_cleaning.ipynb in Jupyter Notebook / VS Code to reproduce the cleaned dataset from the raw CSV.

3. SQL analysis — load cleaned_sales.csv into MySQL Workbench (or your preferred SQL client) and run the queries in SQL/ecommerce_analysis.sql.

4. Dashboard — open PowerBI/Ecommerce_Analytics.pbix in Power BI Desktop to explore the interactive dashboard.

Skills Demonstrated

Python • Pandas • Data Cleaning • Exploratory Data Analysis • SQL • MySQL • Power BI • DAX • Data Visualization • KPI Development • Customer Segmentation • Sales Analysis • Product Analysis • Geographic Analysis • Business Intelligence • Storytelling

Author

Dayyala Manohar Aspiring Data Analyst | Python | SQL | Power BI | Excel

- LinkedIn: <https://www.linkedin.com/in/manohardayyala/>
 - GitHub: <https://github.com/MANOHAR097>
 - Email: manohardayyala1897@gmail.com
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License

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